Pure-Weyl gravity programme: executive summary for physicists

Last substantive update: 18 July 2026.

This is the short, live front door to the programme. It is written for a physicist deciding whether the work intersects their own. It summarizes results; it does not replace the technical manuscripts, certificate ledgers, or the universe-building roadmap.

> **Research context.** This programme is led by Asger Alstrup Palm, Honorary Professor at DTU Compute, as an investigation into whether agentic AI can make auditable contributions to scientific research. Palm is a computer scientist, not a physicist. The physics is therefore offered for expert scrutiny through typed claims, exact symbolic artifacts, independent verifiers, and explicit limitations—not on the authority of either the human director or the AI systems. The DTU affiliation identifies Palm's academic role and does not imply institutional endorsement of the project or its conclusions.

The sixty-second version

Fourth-order gravity has more local solutions than Einstein gravity, and its extra spin-two branch is normally associated with negative energy or norm. Our starting point is that four logically different questions are often collapsed into that one statement:

1. What modes solve the local field equations?
2. Which modes survive gauge, charge, and boundary reduction?
3. Which reduced structures survive interactions?
4. Which survive quantization and define asymptotic particles?

We have exact answers to substantial parts of the first two questions and sharply scoped results on the third. We do not yet have the fourth.

Starting point: choices, not conclusions

The programme began from a deliberately restricted solvable laboratory. The following are inputs to the first calculation, not claims silently promoted to all spacetimes or to the completed quantum theory.

$$S_{\mathrm{W}} = \alpha \int d^4x\, \sqrt{-g}\, C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma}, \qquad M = \mathbb{R} \times S^3.$$

Starting choice	Why it was chosen	What it does **not** imply
Four-dimensional Lorentzian pure-Weyl action	It is the candidate local, conformally invariant fourth-order gravity theory being stress-tested.	That Weyl gravity is correct, stable, unitary, or ghost-free.
Exact conformal cylinder	It is a closed, causal, highly symmetric universe on which the complete gauge complex can be computed exactly.	That the observed universe has this global topology, or that compact results determine boundary charges or scattering.
Vacuum, free, linearized starting theory	It isolates the gravitational constraint, propagation, and pairing problem before adding further failure modes.	Nonlinear closure, generic matter compatibility, black holes, cosmology, or quantum particles.
Diffeomorphisms and local Weyl transformations treated as gauge	They are the defining local redundancies of pure-Weyl gravity.	That every residual conformal transformation is proper gauge on every phase space.
Residual D tested as a constraint in a selected zero-charge sector	It gives an exact test of whether apparent fourth-order modes survive the proposed reduction. Its legitimacy is decided by the covariant charge.	That D is universally gauge. It is charged on the unrestricted compact phase space and must be recomputed with matter or boundaries.

Starting choice	Why it was chosen	What it does **not** imply
The full BV–BFV complex, rather than an isolated fourth-order operator, is the classical object	Fields, ghosts, equations, constraints, identities, causal homotopies, and pairings must be reduced together.	That a raw mode is a physical state, that a Hilbert/Fock space already exists, or that a propagator pole is a particle.

Einstein gravity was not assumed to be a consistent subsector. Constructing its map into the Weyl complex, the relative cofiber, the induced covariant forms, and the mixed nonlinear brackets is now one of the programme's principal tests.

The most complete field-theory result is for free pure-Weyl gravity on the Lorentzian conformal cylinder, $\mathbb{R} \times S^3$, in a selected closed-universe, zero-charge BV–BFV sector. There the full covariant causal complex is connected to the residual state complex by retarded and advanced chain homotopies. After all fifteen residual $SO(4,2)$ generators are imposed as constraints, the vacuum and one-particle residual sectors are acyclic. The first surviving classes are

$$H^4 = \mathrm{span}\{[W_+^2], [W_-^2]\}, \qquad G = I_2.$$

These are positive-paired deformation or vertex classes, not positive-norm graviton particles.

The restriction is physical, not cosmetic. On the unrestricted compact linearized phase space, cylinder time translation D carries a charge and is not gauge. It becomes gauge on the full Taub/moment-map zero fibre and its selected derived quotient. A healthy rotating-scalar Berger-cylinder example shows that a matter clock can coexist with a vanishing total D charge at fixed couplings and linear order, but this is not yet a generic, nonlinear, quantum, or asymptotic theorem.

The honest present verdict is therefore:

> The free classical cylinder theorem survives in its declared sector and now has a covariant causal realization. A universal claim that the fourth-order ghost is absent would be false; interactions, quantum anomalies, extra fourth-order branches, and asymptotic scattering remain decisive.

The newest compact Einstein–Maxwell calculation makes the extra-branch qualification concrete. In the generic axial $\ell \geq 2$ block, the Weyl–Maxwell solution module is strictly larger than the Einstein–Maxwell image by two exact polarizations. The reduced current now matches a direct four-dimensional Weyl–Maxwell Lee–Wald current. The extra block is nonradical with signature $(2, 0)$, while the complete generic axial target has signature $(3, 1)$; unexpectedly, its negative direction lies on one Einstein-image master branch rather than on either extra direction. These are compact classical current signs, not yet quantum norms or particle claims: final residual descent, causal boundary admissibility, and a positive-frequency state space remain open.

The independent polar target operator has now also been reconstructed off shell. On every declared physical $\ell \geq 2$ compact-momentum fibre, including zero momentum, its exact polynomial Einstein square and primary decomposition identify the complete Einstein summand plus two extra polar summands; its action normalization is derived independently from the four-dimensional variation and harmonic norms. The direct polar extra Lee–Wald current, ungauged BV/Noether lift, and final residual descent remain open.

The construction graph below makes the dependency structure explicit. Green nodes are demonstrated in a declared setting, gold nodes are working examples, red nodes record exact limits, and gray nodes are the next frontiers. Every non-open node points back to named machine-verifiable certificates; an arrow means that the lower claim depends on the upper one.

Open the scalable public graph or the complete technical certificate DAG.

How claims are typed

> **Comparison protocol.** Every conclusion is indexed by
>
> `(theory, background, generator, phase space, boundary conditions, lifecycle)`

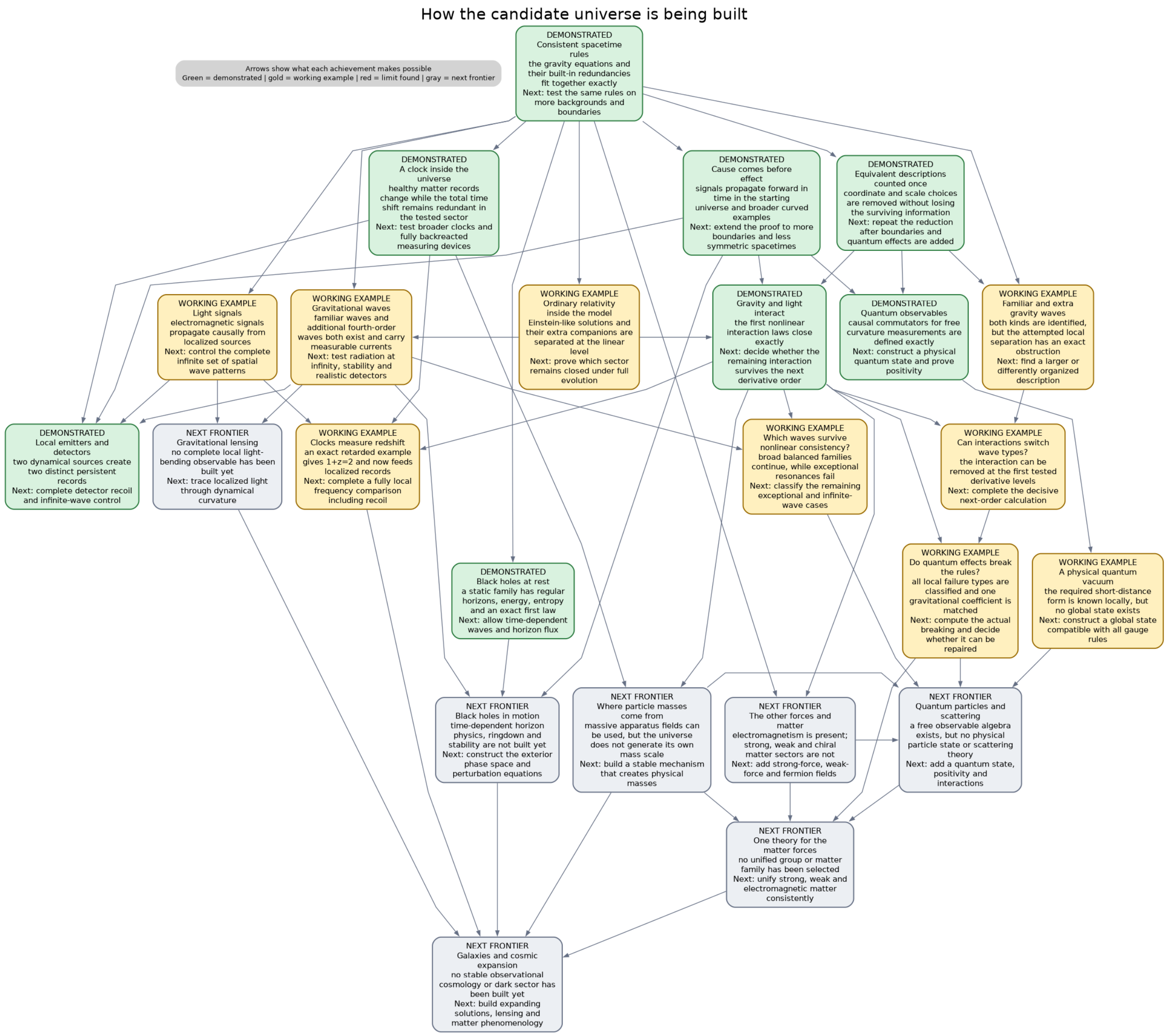

Figure 1: Certificate-backed construction map for the candidate universe

> Two conclusions are compared as competing physical claims only after these six entries have been matched or an explicit map between them has been proved. The same protocol is used in papers, talks, and external bridge notes.

This prevents a local mode, a compact reduced class, a boundary charge, and an asymptotic particle from being treated as four descriptions of the same object without proof. In particular, `LOCAL-ALGEBRAIC` and `REDUCED-MODE` results are not reported as `LORENTZIAN-CAUSAL` or quantum theorems.

We use four plain-language states in this document:

- **Certified**: an exact artifact and an independent verifier establish the scoped claim.
- **Partial**: a real theorem or obstruction is established, but a named dependency needed for the larger conclusion is still open.
- **Fail-closed**: a proposed promotion was tested and is not accepted; this may be a no-go for one ansatz rather than for the theory.
- **Open**: the calculation required for the claim has not been completed.

The result spine

This is the authoritative public claims table. Papers, talks, and bridge notes should link to or quote its scoped rows rather than independently broadening their status.

Question	Current result	Status	Main limitation
Can the free Pais–Uhlenbeck system admit a positive representation?	The positive symplectic metric is reconstructed canonically and its relation to Krein and real-form choices is classified.	**Certified**	Free representation theory is not interacting stability.
Does that positive structure survive interactions?	Explicit resonant conversion channels obstruct analytic continuation of the free positive metric; Einstein–Weyl cubic order is protected but a physical second-order channel is nonzero.	**Certified / Paper 6 under review**	These are specified interactions and sectors, not a universal no-go for every completion.
What is the selected residual cohomology of free pure-Weyl gravity on $\mathbb{R} \times S^3$?	Vacuum and one-particle sectors are acyclic; $H^4 \cong \mathbb{C}^2$, represented by $[W_+^2]$ and $[W_-^2]$, with normalized Gram matrix I_2.	**Certified; Paper 7 artifact-ready**	All fifteen residual generators, including D, are constrained in a zero-charge closed sector. The classes are deformations, not particles.
Is that residual calculation connected to the covariant field theory?	The complete free metric BV–BFV complex has causal retarded/advanced chain homotopies, compact-to-spacelike-compact transport, residual endpoint recovery, and matching Green/current/residual pairings.	**Certified; Paper 8 artifact-ready**	Conformal cylinder, free classical theory, selected polarization; no Hadamard or quantum construction.

Question	Current result	Status	Main limitation
Does causal transfer survive beyond the cylinder?	The abstract cyclic transfer theorem now has complete Berger and flat-Minkowski consumers and a curved Nariai consumer. The repaired Nariai parent-detour cone has 310 components, retracts to the independently certified 26-component metric Bach endpoint, and carries exact advanced/retarded homotopies with the required cyclic adjoint relation.	**Certified on three scoped consumers**	This is one additional curved background, not uniform stability on an open background family or a Hadamard theorem.
Must D be gauge?	No. It is charged on the unrestricted compact phase space and gauge on the Taub-zero derived sector.	**Certified**	The answer is sector- and boundary-dependent.
Can a healthy clock coexist with total D-gauge?	On one positive Berger-cylinder background, fixed-coupling linearized constraints force $\delta Q_D = 0$ although the matter clock has nonzero internal momentum. An exact clock-slice Maxwell observable now evolves nontrivially on that same scoped background.	**Certified in the stated linear/probe sector**	Nearby backgrounds, localized backreacting apparatus, all-orders closure, and quantum stability remain open.

Question	Current result	Status	Main limitation
Is there a clock-defined redshift observable?	An actual retarded Maxwell signal with compact spacetime source support gives a Diff-, Weyl-, Maxwell-gauge-, and raw-D-invariant clock-slice field-strength observable; its exact spatially global fixture has $1+z=2$, a nonzero reduced probe-mode Poisson bracket, and nontrivial clock-time evolution. Separately, two receiver-adjacent localized conserved emitters admit exact causal preparation and produce a leading rank-two detector matrix.	**Global relational observable and localized leading response certified**	The localized preparations are not yet one common emitter experiment or a complete clock-frequency comparison. Evaluated recoil, full apparatus Dirac bracket, all-orders backreaction, phenomenology, and quantum interpretation remain open.
Does the Berger nonlinear Cartan mechanism survive first contact with interactions?	The complete support-local gravitational q_2, q_3, the 64-row gravity–clock–Maxwell q_2, the typed 59,598-term mixed q_3, and the 25,950-term retained ℓ_3 representative are exact and cyclic. The background-preserving $K_{\text{Berger}} = D - \omega R$ Cartan identities close through arity three on the declared classical carrier.	**Certified algebraically and at the stated classical causal level**	This is not an affine raw-D, all-orders, cohomological-nontriviality, or physical branch-mixing theorem. Localized apparatus, QME restoration, and quantum claims remain open.
Does the retained interaction yet distinguish Einstein-like and extra-Weyl branches?	A support-local rank-46 STF2 graph prolongation is an exact cyclic carrier over the retained 36-row complex, with exact SDR and contractible complement. It supplies the enlarged architecture needed to attempt branch resolution.	**Carrier certified; physical split open**	No support-local branch projector or invariant branch manifest has passed. Therefore no Einstein/extra-Weyl/Maxwell ℓ_3 mixing table or physical closure claim is authorized.

Question	Current result	Status	Main limitation
Is ordinary Einstein–Maxwell radiation present inside the Weyl–Maxwell system?	The complete standard fixed-bundle harmonic Einstein–Maxwell tangent injects on shell before the final residual quotient, and its Weyl–Maxwell pullback is nondegenerate.	**Certified, reduced-mode/local-algebraic**	The pullback is not generally the Einstein symplectic form; radiative blocks are relatively indefinite, and the all-sector off-shell relative triangle is incomplete.
Is the complementary fourth-order branch real at the classical level?	For generic compact axial $\ell \geq 2$, the quotient by the Einstein–Maxwell image consists of two exact extra polarizations. Direct four-dimensional Lee–Wald matching makes their block nonradical with signature $(2, 0)$; the full generic axial target has signature $(3, 1)$, with the negative direction on an Einstein-image branch. On every declared physical polar fibre, the independently reconstructed module contains the complete Einstein primary summand plus two extra summands, with action normalization derived from the four-dimensional variation.	**Axial current certified; polar physical module certified**	The polar extra current, ungauged BV/Noether lift, final residual quotient, causal boundaries, positive-frequency state, and particle interpretation remain open.
Is the Einstein sector nonlinearly closed?	Every nonzero pure-extra generic Weyl–Maxwell tangent is Taub-obstructed on the tested fixed compact bundle. Mixed Einstein–extra cones, finite-harmonic and opposite-momentum balanced families, and a two-parameter axial $\ell = 2$ second-order face are classified. The complete exceptional $\ell = 1$, all-m pure cone is also obstructed.	**Pure-extra no-go, scoped mixed extensions, and exceptional all-m no-go certified**	The quadratic source is not disposed on every mixed-cone stratum; generalized/infinite exceptional targets, a structural all-background theorem, and all-orders closure remain open.

Question	Current result	Status	Main limitation
Is there a free Lorentzian quantum-observable algebra?	Gauge-invariant curvature test observables carry an exact causal presymplectic form, and the corresponding curvature-image CCR algebra is defined with the expected causal commutator and quotient relations.	**Certified, LORENTZIAN-CAUSAL free observable algebra**	No compatible positive/Hadamard state, Hilbert representation, interacting product, particle interpretation, or QME theorem follows from the algebra alone.
Can the Berger system support the short-distance structure needed for quantum fields?	The base tensor and ghost wave factors have a certified local Hadamard parametrix, and exact typed Møller intertwiners give the unique formal companion-kernel candidate.	**Partial; microlocal promotion fail-closed**	The order-two transport has no certified Hörmander composition or uniform wavefront control. No companion Hadamard parametrix, global state, QME, or quantum theory is claimed.
Is the quantum theory anomaly-free and unitary?	The even antifield-zero, local dimension-four anomaly candidates reduce to $[\omega C^2]$ and $[\omega E_4]$, with $\omega\Box R$ exact.	**Partial, local-algebraic**	Coefficients, antifield-dependent sectors, QME restoration, Lorentzian time-ordered products, Hadamard state, and asymptotic unitarity are open.
Are black-hole solutions and thermodynamics present?	The static spherical pure-Weyl family is classified in the declared Laurent ansatz, including a non-Einstein regular-horizon fixture. A residual-basic field-dependent generator makes the static charge integrable and gives exact energy, Wald entropy, and $dH = T\,dS$ at every simple horizon.	**Certified static parameter-slice theorem**	No time-dependent horizon phase space, causal exterior problem, flux, ringdown, stability, Hawking state, or general completeness theorem is claimed.

Highlights by audience

Higher-derivative quantization and unitarity

The programme separates the free positive metric, Krein completion, real form, gauge reduction, and interacting deformation problem. The mechanical and flat-field papers show why a free positive representation can be both mathematically canonical and physically insufficient. The gravity papers then ask whether gauge reduction changes the carrier space before positivity is judged.

The potentially useful contribution is a common diagnostic framework for claims that a ghost is removed, rendered null, or made positive. Those claims can disagree because they use different observable algebras, phase spaces, boundary conditions, quotients, or inner products. Our cylinder result does not settle the Mannheim versus

Fock–BRST debate; it supplies a third, cohomological reduction whose state type must not be confused with an asymptotic Fock particle.

The key open comparison is one common flat or asymptotic fixture evaluated in all three descriptions.

Possible novelty, pending literature audit: a claim-typed comparison protocol that separates Fock states, BV classes, deformation classes, boundary charges, and asymptotic particles before competing ghost or unitarity claims are compared.

BV, BRST, and perturbative quantum field theory

Papers 7–8 give an explicit large gauge complex rather than a mode-only quotient. The residual BFV complex is obtained from the metric BV complex, the quadratic moment map is identified with the Taub obstruction, and the derived zero fibre explains why the zero-charge condition cannot be omitted.

For quantum field theorists the present result is a classical input, not a QME theorem. The immediate target is to finish local anomaly cohomology, compute the actual type-A and type-B coefficients by two independent methods, restore or obstruct the QME, and only then ask whether the D-Cartan identity transfers quantum mechanically.

Possible novelty, pending literature audit: a coefficient-bearing map from the standard type-A/type-B Weyl-anomaly classes to the obstruction of a specific residual Cartan contraction, after the full antifield and QME gates are closed.

Matter content, scale generation, and unification

The programme does not yet contain a Grand Unified Theory. A GUT would unify the strong, weak, and electromagnetic gauge sectors; it would not by itself quantize or unify gravity. The relevant long-range target here is an anomaly-free unified gauge–matter sector consistently coupled to pure-Weyl gravity.

The first credible bridge result would be smaller: construct a non-Abelian Yang–Mills/chiral-fermion BV complex, classify gauge, mixed gauge–gravity, and Weyl anomalies, and determine whether their cancellation selects matter representations. A GUT candidate additionally requires a conformal mass-generation and symmetry-breaking mechanism, a physical particle spectrum, running couplings, and low-energy recovery. Likewise, $E = mc^2$ is not an independent construction target; it follows at rest from a Lorentz-invariant mass shell once a stable physical mass scale and massive excitation exist.

Lorentzian PDE and Green-hyperbolic complexes

The free pure-Weyl metric Hessian does not become a scalar normally hyperbolic operator on a simple same-bundle auxiliary enlargement; exact symbol-rank no-go results explain that failure. The successful object is the whole complex. Curvature prolongation, a symmetric-hyperbolic Weyl–Cotton system, adjoint-tractor transfer, and support-local retracts produce retarded/advanced degree-minus-one homotopies on all 386 prolonged rows.

This construction has now been extracted into a conditional cyclic causal-transfer theorem. A finite-order support-local cyclic strong deformation retract transfers same-sided advanced/retarded chain homotopies by $\Lambda_C = h + i\Lambda_E p$, preserves their causal support and cyclic adjoint relation, and is stable under finite direct sums and finite-order cyclic shears. The complete Berger 54- and 64-row complexes replay as exact consumers. Endpoint Green hyperbolicity remains a hypothesis rather than a conclusion of the theorem.

Possible novelty, pending literature audit: constructive cyclic transfer of retarded/advanced Green homotopies and current pairings from a hyperbolic parent to a fourth-order gauge/detour complex through support-local differential contractions.

On the Berger quantum route, the normally hyperbolic tensor and ghost factors now carry the standard local Hadamard singularity, and the typed source and solution Møller maps satisfy their exact formal intertwining and adjoint identities. The attempted promotion usefully found the remaining analytic obstruction: convergence in finite-slab Sobolev energy norms does not imply that the kernel compositions are defined in the Hörmander sense or that their wavefront cones are uniformly controlled. Thus the formal transported kernel is fixed, while the companion Hadamard claim remains fail-closed.

The newest contraction theorem sharpens the remaining work: a valid covariance on the retained 26-row complex would lift canonically to all 54 rows without new wavefront directions or an independent state on the 28 contractible rows. The stationary-pencil preflight has now converted the retained equations into an exact rank-52 second-order hybrid companion and a rank-104 first-order Cauchy target. The next gate is a closed realization of that generator with an isolated zero spectral sector; no Riesz projector, covariance, or Hadamard state has yet been constructed.

Conformal geometry, tractors, BGG, and detour complexes

The geometric bridge uses the adjoint-tractor Yang–Mills detour complex and curved BGG/homological transfer to reach the metric Bach complex. This gives Lorentzian causal content to structures usually presented through formal self-adjointness, ellipticity, or representation theory.

The result of interest here is not merely another realization of the Bach operator. It is that the reduced fourth-order operator can inherit causal meaning from a larger tractor complex even when a preferred scalar-principal factorization of the isolated metric operator does not exist. A nearby conformal higher-spin or detour example is the natural portability test.

Possible novelty, pending literature audit: a Lorentzian causal interpretation of a Bach detour complex inherited from its tractor parent, including explicit differential splitting, homotopy, support, and pairing transport rather than formal self-adjointness alone.

General relativity and mathematical relativity

The compact Einstein–Maxwell work separates three statements that are often conflated:

1. Einstein solutions solve the Weyl equations on a common background.
2. Einstein linear tangents inject into the Weyl solution space.
3. The Einstein sector is preserved by nonlinear evolution and has the same symplectic form.

The first two now hold in the complete certified standard harmonic tangent before the final quotient. The third does not follow: the Weyl–Maxwell pullback is nondegenerate but differs blockwise from the Einstein–Maxwell form, and fixed-charge second-order Taub obstructions occur for explicit photon and gravitational modes.

The complementary axial block is now explicit rather than inferred from a determinant. For every physical $\ell \geq 2$ in the generic compact domain, the quotient contains two extra solution polarizations. Their reduced current has now been matched to the direct four-dimensional Lee–Wald current. The extra block is nonradical with signature $(2, 0)$, but the complete axial target has signature $(3, 1)$: the negative direction occurs on one Einstein-image master branch. This rules out both the claim that the extra root is merely a radical multiplicity and the naive identification of the extra root with the negative direction on this fixture. Final residual, positive-frequency, and causal-boundary selection still decide the physical interpretation.

The polar block has now crossed the physical coefficient-module gate. Independent four-dimensional linearizations reconstruct its formally self-adjoint four-by-four Hessian. For every declared physical $\ell \geq 2$ and compact momentum, including zero, the exact polynomial square identifies the complete Einstein primary image and two additional polar primary summands. The action row weights are derived from the four-dimensional variational convention and harmonic norms. This is not yet the polar analogue of the preceding physical-current theorem: its direct Lee–Wald form and final gauge/residual descent remain to be computed.

At second order, the latest real axial–polar $\ell = 2$ fixed-bundle tangent has a positive-definite Taub form and is obstructed in every nonzero real direction. A separately computed sum-frequency source is removable; the obstruction instead comes from conjugate self-products in the zero-frequency constraint channel. This distinction is why individual vertex blocks cannot substitute for the full linearization-stability test.

This connects most directly to linearization stability, charge-fibre obstructions, and the Lorentzian complement to boundary-selected "Einstein from conformal gravity." The decisive next step is the full extra fourth-order quotient and then an asymptotically flat Bach phase space with Bondi/ADM charges and flux.

Possible novelty, pending literature audit: explicit dependence of second-order integrability on the global charge fibre, together with a harmonic obstruction bilinear that separates removable dynamical sources from Taub/cokernel obstructions.

Relational clocks and the problem of time

The charge calculation replaces the slogan "time is gauge" with a test:

$$\delta H_D = \Omega(\delta\phi, \mathcal{L}_D\phi).$$

For unrestricted compact data this is nonzero. On the Taub-zero sector it vanishes. The Berger construction adds a healthy rotating scalar reference system and shows, at fixed couplings and linear order, that nontrivial matter clock momentum need not make the total D transformation physical.

The first explicit observable now uses an actual retarded Maxwell response to a source compact in spacetime. The clock-slice field-strength observable is Diff-, Weyl-, Maxwell-gauge-, and raw-D-invariant, has a nonzero reduced probe-mode Poisson bracket and evolves nontrivially; its exact spatially global fixture has $1 + z = 2$. The clock-dressed source schedule is correctly an equivariant family, not an invariant source at fixed schedule. This is a G0 retarded relational-observable theorem; it is spatially global. A separate observer theorem now constructs two receiver-adjacent localized conserved emitters, exact causal preparations, and a leading rank-two detector matrix. It does not yet identify those preparations with one common emission event or the complete clock-frequency observable. The full apparatus Dirac bracket, evaluated recoil, all-orders backreaction, and phenomenology remain open.

The first gravity-coupled stress projection now sharpens that gate. The diagonal energy-pressure source is exact, but a single travelling Hopf mode has nontrivial stationary momentum flux with an exact dual obstruction witness. A coherent counter-propagating mode in the same Maxwell field forms a standing wave, cancels that flux including interference stress, and admits an explicit second-order homogeneous gravity correction. Its phase plane has positive energy and no new negative direction. The next physical test is to join the localized preparations into one emitter–receiver comparison and evaluate apparatus recoil or a support-local response to the lone travelling beam—not another homogeneous balance calculation.

Possible novelty, pending literature audit: an explicit healthy clock with nontrivial relational evolution while total D remains presymplectically degenerate, extended to a gauge-invariant light observable and its first interaction obstruction.

Amplitudes, twistors, and Einstein from conformal gravity

Our current Einstein result is complementary to boundary selection and twistor embeddings: it asks whether the selected Einstein tangent is a causally and symplectically closed sector, and whether it extends beyond linear order. It does not yet compute an S-matrix.

A useful bridge would identify the certified Einstein projection in helicity or twistor variables and evaluate one cubic or MHV fixture. A later result should determine whether the compact charge-sector obstruction has an amplitude or soft-charge interpretation.

Phenomenology, black holes, and cosmology

This programme is not yet a dark-matter, dark-energy, or dynamical black-hole model. It now has a classified static spherical pure-Weyl family, a regular non-Einstein horizon fixture, and an exact normalized static first law. The time-dependent horizon phase space, flux, ringdown and stability gates remain open. Galaxy and cosmology applications likewise require stable backgrounds, observables, and boundary charges before fitting rotation curves or expansion histories is meaningful.

The infrastructure is relevant because it can test whether the extra Weyl branch that drives such phenomenology is physical, gauge, constrained, unstable, or boundary-selected. The sequence is: asymptotic and horizon phase spaces; extra-branch classification; nonlinear stability; redshift, lensing, and waveform observables; only then galaxy and cosmology fits.

What we can offer collaborators now

The programme is not only a collection of conclusions. A collaborator can currently use:

- exact coefficient-level operators, pairings, contractions, currents, and obstruction witnesses with declared conventions and provenance;
- small independent verifiers and machine-readable certificates rather than requiring trust in the main symbolic producer;

- compact cylinder, Berger-clock, Einstein–Maxwell product, Nariai, and static spherical black-hole backgrounds as reproducible test laboratories;
- explicit missing-object and failed-promotion ledgers, including normalized defects that can be attacked without adopting the programme's interpretation;
- tractor/BGG, BV, Green-homotopy, Lee–Wald, Taub, and harmonic adapters that can receive an outsider's preferred formulation or benchmark.

The practical offer is: provide an operator, background, boundary condition, mode, charge convention, or proposed reduction, and we can attempt to place it in the comparison protocol and run it through the relevant certificate chain.

Bridge-project status

Readiness is reported by lifecycle and missing gate.

Bridge project	Current readiness	Exact next gate
Cyclic causal Green transfer	**Abstract conditional theorem certified** with Berger, flat-Minkowski, and curved Nariai consumers; the latter closes all 310 repaired rows	Formulate and prove a uniform open-background (`G3`) version or add a different detour/higher-spin theory.
Pure-extra obstruction and balanced extension	**Theorem frozen** on the generic fixed-bundle domain; finite-harmonic, opposite-momentum, and exceptional all-m successors certified	Dispose the phase-sensitive quadratic source on the remaining mixed strata and extract the structural definite/indefinite Taub theorem.
Extra axial branch and physical current	**Axial direct current and polar physical module certified**	Compute the polar extra Lee–Wald current and ungauged lift, perform final residual descent, and test causal boundary admissibility.
Relational clock and light	**Spatially global retarded redshift plus localized causal emitters and leading rank-two detector response certified**	Unite the local preparations into one clock-frequency comparison and complete recoil, backreaction, and the apparatus Dirac bracket.
Static black holes	**Background family, horizon-regular fixture, normalized energy, entropy, and exact static first law certified**	Construct the time-dependent horizon phase space and causal exterior perturbation theory before ringdown or stability claims.
Residual branch mixing	**Rank-46 cyclic carrier certified; retained ℓ_3 representative certified**	Construct or obstruct a support-local branch projector, then compute the background-indexed Einstein-like/extra-Weyl/Maxwell mixing table and its invariant meaning.
Quantum anomaly bridge	**Candidate classes classified** in the stated antifield-zero sector	Compute coefficients by two methods, complete antifields, and restore or obstruct the QME and D identity.
Asymptotic Bach/BMS	**Programme stage**	Construct a closed Lorentzian boundary phase space with differentiable charges, flux, and extra-branch signs.

The charge-fibre project has been promoted from a bridge note to a standalone scoped manuscript: it proves the complete pure-extra generic obstruction and one explicit balanced Einstein–extra second-order extension. The cyclic and extra-current bridge notes retain their shorter adapter format.

Where the strongest criticism currently lands

The criticism is correct in four important senses:

- D is not universally gauge. It is charged on the unrestricted compact phase space and is expected to be physical in ordinary asymptotic settings.
- A zero one-particle residual cohomology on the selected closed cylinder is not a proof of particle unitarity or of the absence of radiative degrees of freedom in an asymptotically flat universe.
- The Einstein image is not automatically a nonlinear, symplectic, or exclusive sector of the fourth-order theory.
- In the generic compact axial block, the extra fourth-order module survives both the reduced Green-pairing radical test and the direct four-dimensional Lee–Wald comparison. It can no longer be dismissed there as determinant multiplicity, although its residual, boundary, and quantum admissibility remain undecided.
- In the Berger interaction theorem, the exact nonlinear Cartan generator is $K_{\mathrm{Berger}} = D - \omega R$, not affine raw cylinder translation D. Raw-D nonlinear closure therefore remains open.

The selected construction nevertheless survives the criticism where it actually claims to apply:

- the zero-charge condition is now derived and audited rather than assumed silently;
- the residual calculation is connected to the complete covariant causal free complex;
- a healthy clock counterexample shows that matter does not automatically turn total D into a charge;
- the nonlinear Berger K_{Berger}-Cartan recurrence closes through arity three; mixed gravity–clock–Maxwell q_2, q_3 and the retained ℓ_3 representative are certified; and an actual retarded relational signal exists. Localized emitter preparations and a leading rank-two response now exist as a separate observer theorem. The branch projector, invariant interaction meaning, unified clock-frequency comparison, recoil, and complete backreacted apparatus remain open.

Thus the sector is neither an arbitrary patch nor a model of the whole universe. It is a mathematically consistent physical choice whose range of stability is now the subject of the programme.

What would materially change the verdict

A strong positive change would be any of:

- a single localized Berger emission-and-reception experiment whose complete clock-frequency comparison, recoil, backreaction, and apparatus Dirac bracket preserve the certified leading rank-two response;
- a support-local rank-46 branch projector followed by a nonremovable Einstein-like/extra-Weyl/Maxwell interaction coefficient;
- localized apparatus recoil or a support-local single-beam response compatible with the arity-three Cartan contraction;
- a polar extra Lee–Wald current, ungauged lift, and final residual descent that either preserve or sharply obstruct the axial classification under causal physical boundary conditions;
- vanishing or removable quantum D-anomaly after QME restoration;
- an asymptotically flat, positive reduced Einstein scattering sector with the extra Weyl branch excluded or controlled.

A strong negative change would be any of:

- failure of the Berger contraction in the next physical interaction channel;
- survival of the compact negative Lee–Wald direction as an unavoidable physical state after residual descent and admissible causal boundaries;
- an unavoidable nontrivial quantum Cartan anomaly;
- a negative physical extra branch in the asymptotic Bach phase space;
- failure of causal nonlinear closure for every physically admissible Einstein-sector boundary condition.

Either outcome is publishable: the programme is designed to locate the first precise obstruction, not to protect a preferred interpretation.

Reading and verification map

- Series overview and papers
- Paper 7: residual cohomology
- Paper 8: covariant causal transport
- Papers 7–8 computational supplement
- Paper 9: Berger relational clocks and Cartan contraction
- Paper 10: compact Einstein–Maxwell/Weyl–Maxwell phase space
- Clean publication-release audit
- Live D-quotient status ledger
- Generic axial extra-branch and pairing report
- Generic polar off-shell operator certificate
- Physical polar module and action-normalization certificate
- Complete Berger gravity–clock–Maxwell q_2 report
- Real axial–polar Taub obstruction receipt
- Berger arity-three Cartan report
- Berger Maxwell interaction-obstruction report
- Berger momentum-balanced second-order fixture
- Retarded Berger relational Maxwell observable
- Localized dynamical-emitter rank-two response
- Curved Nariai 310-row causal transfer
- Free curvature-observable causal propagator
- Free curvature-image CCR algebra
- Static pure-Weyl black-hole background
- Normalized black-hole generator and first law
- Rank-46 cyclic Berger branch carrier
- Paper 11: retained mixed gravity–Maxwell bracket
- Berger typed Møller and microlocal-gate report
- Bridge note: cyclic causal Green transfer
- Paper 91: pure-extra obstruction and balanced extension
- Exceptional all-m $\ell = 1$ obstruction
- Bridge note: axial current and polar module audits
- Long-term programme and publication gates
- General-audience introduction

Papers 7–8 are `ARTIFACT_READY`: the manuscripts, supplements, hashes, and clean reproduction audit pass. Their next lifecycle gate is a public repository release audit, including link integrity, provenance, readable scope boundaries, and a clean tagged replay.

Papers 9–10 have frozen scoped theorems. Paper 11 is a working open-repository paper with a certified retained representative; its narrower theorem can be frozen independently of the still-open physical branch and deformation-class questions.

Update policy

This document should remain short enough to read in roughly ten minutes. A new calculation changes it only when at least one of the following happens:

1. a lifecycle state advances or retreats;
2. a theorem gains or loses a dependency needed for its physical reading;
3. a new background or boundary condition changes a verdict;
4. an open decisive test becomes a certified pass, obstruction, or no-go;
5. a manuscript becomes artifact-ready or repository-released.

Raw term counts, implementation milestones, and unverified candidate results belong in team reports, not in this front door. Every update should preserve the strongest limitation immediately beside the headline result.

Changelog

- **18 July 2026, curved-causal, observer, black-hole, and quantum-observable update:** added the complete 310-row Nariai causal consumer, localized dynamical-emitter rank-two response, exceptional all-m $\ell = 1$ obstruction, the free curvature-observable CCR algebra, and the static pure-Weyl black-hole family with its normalized exact first law; retained the open-background, full-apparatus, quantum-state, and dynamical-horizon boundaries.
- **17 July 2026, observable and branch-carrier update:** promoted the Berger redshift fixture to an actual retarded spatially global relational observable; recorded the landed mixed q_3, retained ℓ_3, and exact rank-46 cyclic carrier; kept localization, branch projection, invariant interaction meaning, and apparatus backreaction explicitly open.
- **17 July 2026, relative-spine update:** corrected the Berger nonlinear generator from affine raw D to $K_{\mathrm{Berger}} = D - \omega R$; added the complete 64-row gravity–clock–Maxwell q_2, the all-physical-fibre polar module and action normalization, and the sectoral Einstein–Weyl relative-complex status.
- **17 July 2026:** promoted the Berger Cartan result through arity three; recorded the one-way Maxwell momentum-flux obstruction and its exact counter-propagating second-order resolution; replaced the pending axial Lee–Wald gate with the completed direct-current match, signatures, and real axial–polar Taub obstruction; added the comparison protocol, collaborator offer, possible-novelty statements, and exact bridge gates.
- **16 July 2026:** added the generic axial strict-inclusion theorem and its nonradical reduced Green signature $(2, 0)$; recorded the exact typed Møller transport algebra and the fail-closed Hörmander/wavefront gate that prevents promotion to a companion Hadamard parametrix or quantum state.
- **16 July 2026:** created the physicist-facing summary; recorded Papers 7–8 as artifact-ready, the completed cylinder causal bridge, the sector-dependent D verdict, Berger arity-two status with the separate fail-closed analytic import, and the complete standard Einstein–Maxwell harmonic inclusion before the final quotient.